GENERAL ARRANGEMENT DRAWING

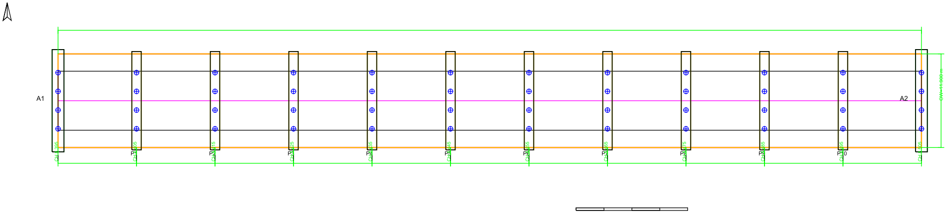
11 × 10 m RCC Solid Slab Major Bridge · 110 m · GAD-MB-11SPAN-SLAB-001

7 coordinated sheet(s)

IRC / MoRTH presentation booklet

Not a construction IFC issue — GAD for review

Sheet 1/7 — Sheet1 Plan

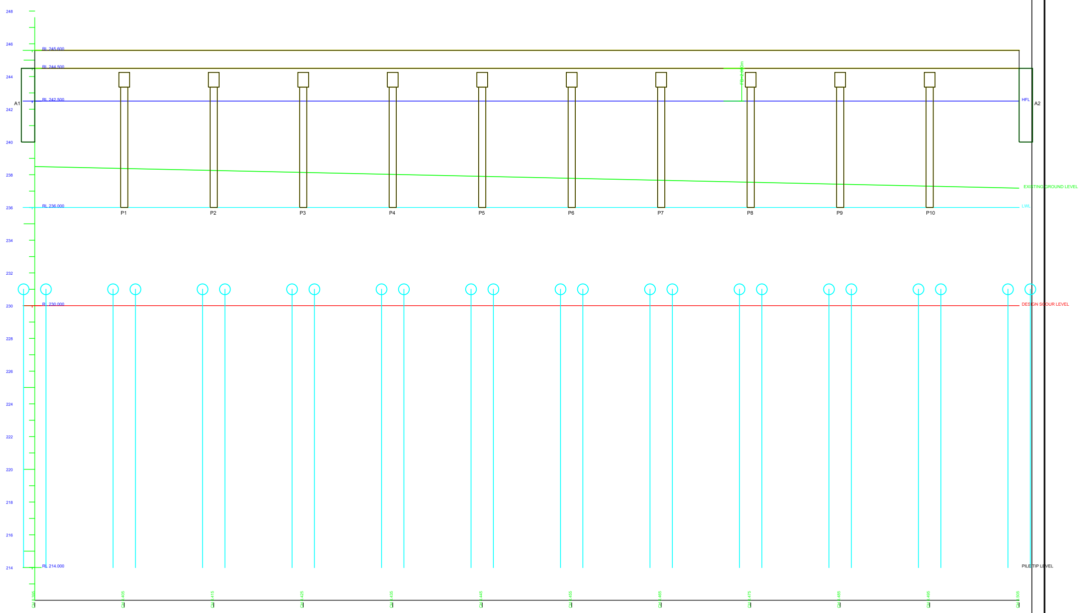


MAJOR-BRIDGE-CH8-450
MAJOR BRIDGE ACROSS RIVER PATHERI AT CH 8+450 PMGSY PKG-III

GENERAL ARRANGEMENT — PLAN VIEW
Consultant: TIER1_ENGINEERING
Scale: 1:200 | Client: STATE_PWD
Designed: Er. A. Deshpande | Checked: Er. P. Joshi | Approved: Er. S. Patil (CE — PMGSY)
DRG: GAD-MB-11SPAN-SLAB-001 | Rev: R1 | Date: 2026-10-05 | Sheet 1 / 7

PLAN VIEW (indicative 1:200)

Sheet 2/7 — Sheet2 LongSection



LONGITUDINAL SECTION

MAJOR-BRIDGE-CH8-450
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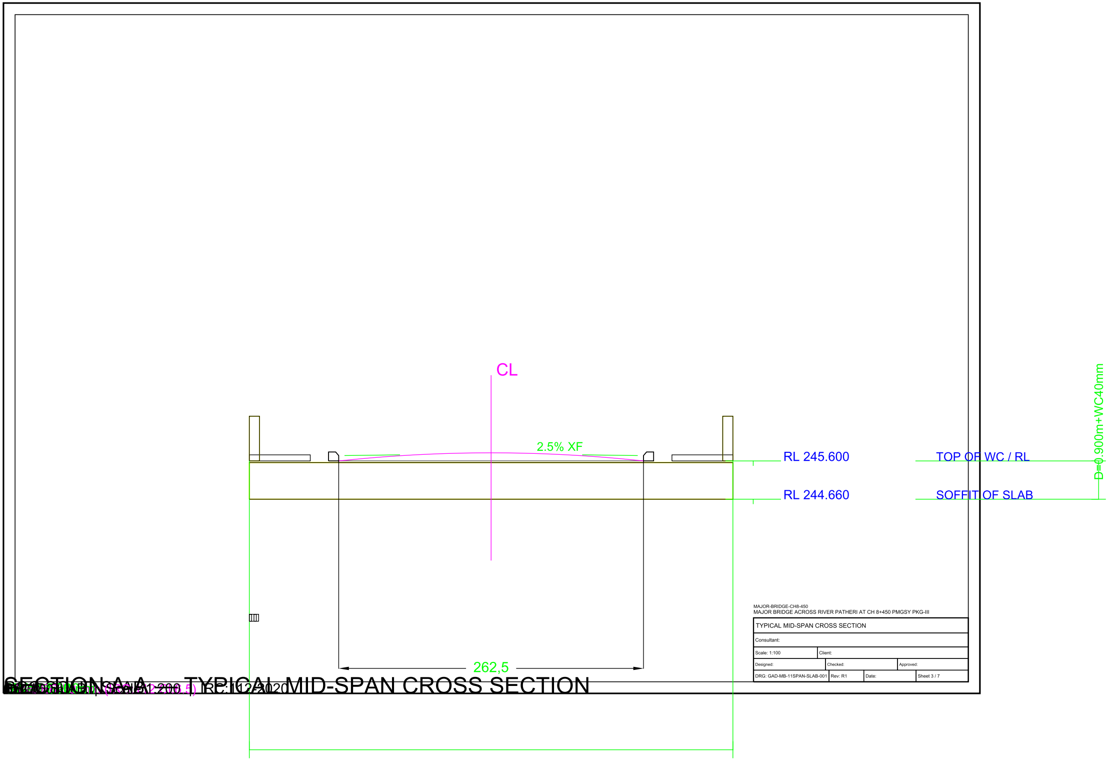
GENERAL ARRANGEMENT — LONGITUDINAL SECTION			
Consultant: TIER1_ENGINEERING			
Scale: 1:200 H 1:100 V	Client: STATE_PWD		
Designed: Er. A. Deshpande	Checked: Er. P. Joshi	Approved: Er. S. Patil (CE — PMGSY)	
DRG: GAD-MB-11SPAN-SLAB-001	Rev: R1	Date: 2026-10-05	Sheet 2 / 7

Sheet 3/7 — Sheet3 CrossSection



MAJOR-BRIDGE-CH8-450
MAJOR BRIDGE ACROSS RIVER PATHERI AT CH 8+450 PMGSY PKG-III

TYPICAL MID-SPAN CROSS SECTION

Consultant:

Scale: 1:100
Client:

Designed:
Checked:
Approved:

DRG: GAD-MB-11SPAN-SLAB-001
Rev: R1
Date:
Sheet 3 / 7

Sheet 4/7 — Sheet4 Foundation

PARAMETER	VALUE	UNIT
Pile Type	BORED_CAST_IN_SITU	
Pile Diameter	1200	mm
Pile Length	18.0	m
Piles per Pier	4	nos
Spacing c/c	2.5	m
Cap Thickness	1500	mm
Pile Cutoff RL	232.0	m
FOS Bearing	3.0	-
Allow. Bearing Cap.	450.0	kPa
Scour Level RL	230.0	m
Scour Depth	3.5	m

PILE SCHEDULE

MAJOR-BRIDGE-CH8-450
MAJOR BRIDGE ACROSS RIVER PATHERI AT CH 8+450 PMGSY PKG-III

FOUNDATION DETAILS — TYPICAL PIER			
Consultant:			
Scale: 1:100	Client:		
Designed:	Checked:	Approved:	
DRG: GAD-MB-11SPAN-SLAB-001	Rev: R1	Date:	Sheet 4 / 7

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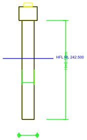
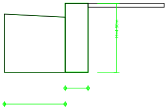
11SPAN_110m_RCC_SLAB_GAD

Sheet 5/7 — Sheet5 PierAbutment





1. All dimensions in metres unless stated.
2. Concrete grade pier/abut: M25.
3. Backfill: SELECT GRANULAR MOORUM COMPACTED.
4. Approach slab reinforced with Fe500D bars @ 150 c/c both ways.
5. All exposed surfaces to have 2-coat epoxy waterproofing.

MAJOR-BRIDGE-CH8-450
MAJOR BRIDGE ACROSS RIVER PATHERI AT CH 8+450 PMGSY PKG-III

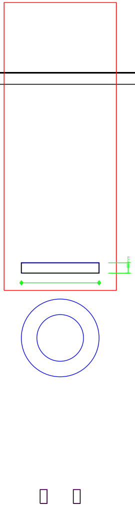
PIER AND ABUTMENT DETAILS			
Consultant:			
Scale: 1:100	Client:		
Designed:	Checked:	Approved:	
DRG: GAD-MB-11SPAN-SLAB-001	Rev: R1	Date:	Sheet 5 / 7

[illegible]

Sheet 6/7 — Sheet6 BearingsJoints

BEARING SCHEDULE (IRC:83)

Location	Type	Size X (mm)	Size Y (mm)	Load (kN)	Movement	Layers	Qty
A1-ELAST-FIX	ELASTOMERIC_	400	400	1200.0	FIXED		2
P1-P10-ELAST	ELASTOMERIC_	400	450	1800.0	FREE		40
A2-ELAST-FRE	ELASTOMERIC_	400	400	1200.0	FREE		2



EXPANSION JOINT SCHEDULE (IRC SP:55)

Location	Type	Width (mm)	Movement (±mm)	Gap (mm)	Length (m)	Qty
A1-End Singl	SINGLE_MODUL	120	60	30	12.0	1
A2-End Singl	SINGLE_MODUL	120	60	30	12.0	1

BEARING & EXPANSION JOINT NOTES:

1. Bearings shall conform to IRC:83 (Parts I, II, III) as applicable.
2. Elastomeric bearings: shore hardness 60±5, compression set ≤25%.
3. POT bearings: pot pressure ≤35 MPa; PTFE sliding surface Ra≤0.4μm.
4. Strip seal joints: neoprene seal ASTM D2000; anchor steel S355JR HDG.
5. Bearing replacement provision: jacking sockets at each bearing seat.
6. Inspect bearings every 5 years; replace elastomeric pads at 25 years.

MAJOR-BRIDGE-CH8-450
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BEARINGS AND EXPANSION JOINTS SCHEDULE			
Consultant:			
Scale: 1:100	Client:		
Designed:	Checked:	Approved:	
DRG: GAD-MB-11SPAN-SLAB-001	Rev: R1	Date:	Sheet 6 / 7

[illegible] — PLAN

MB-11SPAN-SLAB_Sheet6_BearingsJoints.dxf   ent=167 ren=167 skip=0
11SPAN_110m_RCC_SLAB_GAD

Sheet 7/7 — Sheet7 BOQ

Chainage: 8.450 km | State: MH | Total Length: 110.000 m | Overall Width: 11.900 m

Item	Description	Unit	Quantity	Rate (INR)	Amount (INR)	Ref.
1	EARTHWORK — EMBANKMENT (IMPO	CUM	3272.5	TBC		MORTH 300
2	GRANULAR SUB-BASE (GSB) — AP	CUM	206.2	TBC		MORTH 401
3	CONCRETE M25 — PILE FOUNDATI	CUM	480.00	TBC		MORTH 1700
4	CONCRETE M30 — SUBSTRUCTURE	CUM	620.00	TBC		MORTH 1700
5	CONCRETE M40 — SUPERSTRUCTUR	CUM	1153.00	TBC		MORTH 1700
6	REINFORCEMENT STEEL Fe500D —	MT	165.00	TBC		MORTH 1600
7	PRESTRESSING STEEL (STRANDS	MT	0.00	TBC		MORTH 1600
8	FORMWORK — SUPERSTRUCTURE	SQM	5188.5	TBC		MORTH 1700
9	BITUMINOUS CONCRETE (BC) WEA	SQM	825.0	TBC		MORTH 510
10	ELASTOMERIC / POT BEARINGS (	EACH	44	TBC		MORTH 2800
11	EXPANSION JOINTS (STRIP SEAL	RM	23.8	TBC		MORTH 2800
12	APPROACH SLAB M30 (BOTH ENDS	CUM	29.75	TBC		MORTH 1700
13	RIVER TRAINING WORKS (LUMP S	LS	1	TBC		MORTH 2500
14	RAILING / CRASH BARRIER (M.S	RM	240.0	TBC		MORTH 800
15	WATERPROOFING MEMBRANE — DEC	SQM	1309.0	TBC		MORTH 2600
	TOTAL (CIVIL WORKS — BRIDGE)		**TBD**	TBC		

NOTES:

1. Quantities are preliminary estimates for budget purposes only.
2. Final quantities to be measured as per IS:1200 / MORTH specifications.
3. Rates as per prevailing State PWD / MORTH Schedule of Rates.
4. 15% contingency to be added for unforeseen items.
5. Preliminary estimated cost: To Be Determined (TBD).

MAJOR-BRIDGE-CH8-450
MAJOR BRIDGE ACROSS RIVER PATHERI AT CH 8+450 PMGSY PKG-III

ABSTRACT BILL OF QUANTITIES			
Consultant:			
Scale: 1:100	Client:		
Designed:	Checked:	Approved:	
DRG: GAD-MB-11SPAN-SLAB-001	Rev: R1	Date:	Sheet 7 / 7

ABSTRACT OF ESTIMATED COST — MAJOR-BRIDGE-CH8-450